Academic Curriculum Vitae

L. Janindu Nethmal Peiris

Academic Profile

A self-motivated and passionate individual aspiring to pursue graduate studies abroad and contribute to advanced research in Mathematics. I recently served as a Teaching Assistant in the Department of Mathematics, University of Sri Jayewardenepura, Sri Lanka. I possess a strong academic background with deep interests in Mathematical Biology, Functional Analysis, Topology, Computational Mathematics and Numerical Analysis. I initially pursued the subject combination of Mathematics, Statistics, and Computer Science, qualifying to follow a honors degree in any of these three disciplines, and ultimately chose to specialize in Mathematics. Having consistently demonstrated academic excellence, I was honored as the highest GPA holder in Mathematics in my batch, which initially comprised over 350 students.

In the last semester, I also worked as the Lecturer in Charge of MAT 1100 Mathematical Software at the Department of Mathematics, University of Sri Jayewardenepura. In this role, I was responsible for delivering lectures on essential mathematical software tools and computational techniques, with a primary focus on Maple. My duties further included preparing, evaluating, and discussing assignments, exercises, and the end-semester examination. Additionally, I oversaw tutorial sessions by instructing and guiding students, thereby enhancing their understanding and overall learning experience.

Personal Information

Name:	LJN Peiris (Labugamaralalage Janindu Nethmal Peiris)
Current Role:	Visiting Lecturer (GISM - Graduate Institute of Science and Management)
Date of Birth:	July 17, 2000
Marital Status:	Unmarried
Gender:	Male

Contact Information

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Education

- **University of Sri Jayewardenepura, Nugegoda, Colombo, Sri Lanka**
  June 2021 – April 2025 (Graduated: 17th of October 2025)
  - ⋆ B.Sc. (Hons) in Mathematics — First Class
    Overall GPA: 3.8115 / 4.00
    Subject-wise GPA: Mathematics — 3.94 (80 credits), Statistics — 3.74 (20 credits), Computer Science — 3.35 (20 credits)
    - ◇ Thesis Topic: An Investigation of Fredholm Operators and Fredholm Index on Separable Hilbert Spaces.
    - ◇ Supervisors: Dr. R.P.K. De Silva and Mr. K. K. W. A. Sarath Kumara
  - ⋆ Followed the subject combination — Mathematics, Statistics, and Computer Science
    Duration: June 2021 – May 2023
    Qualified to pursue a special degree in any of the three disciplines and ultimately chose to specialize in Mathematics.

  Scan above QR code or Click here to view my official final results

- **Prince of Wales' College, Moratuwa, Sri Lanka** **2011 – 2019**
  - ⋆ G.C.E. Advanced Level Examination, August 2019 (Mathematics Stream – Combined Mathematics, Chemistry, Physics; Results: 3 B's; Z-score: 1.546)
  - ⋆ G.C.E. Ordinary Level Examination, 2016 December (9 A's)

- **Mampe Primary School, Piliyandala, Sri Lanka** **2005 – 2010**
  - ⋆ Fifth Grade Scholarship Examination – 2010 August: Passed with a score of 158 (Cut-off: 148)

Honors and Awards

- **Prof. Sunethra Weerakoon Gold Medal** – awarded to the student who obtains a First Class Honours degree with the best performance in Mathematics or Applied Mathematics, recognizing the highest GPA in the specialized subject area within the Department of Mathematics. (Received at the 51st Convocation of the University of Sri Jayewardenepura, 17th of October 2025)
- **Dr. Sirimathi Wewala Memorial Scholarship** for obtaining the highest performance (GPA) in the first three years of the B.Sc.Mathematics (Special) degree at the University of Sri Jayewardenepura, Department of Mathematics. (July 2024)
- **Three-time Gold Medalist** – Annual Prize Giving, Prince of Wales' College:
  - 2016 – Grade 11 Batch First
  - 2017 – Grade 12 Mathematics Batch First
  - 2018 – Grade 13 Science Batch First

Academic Fellowships and International Programs

- **Member of the Manta Institute of Mathematics**, participating in academic programs and collaborating with a global network of mathematicians.
- **Math in Moscow Program** — Selected as one of 13 Manta Fellows from three leading state universities, Sri Lanka, fully funded by the Manta Institute for Mathematics, to participate in the prestigious Math in Moscow program which commenced in September 2025.

  Organized by the Independent University of Moscow, the Higher School of Economics, and the Moscow Center for Continuous Mathematical Education, this highly competitive international program brings together exceptional mathematics students from across the globe for advanced coursework, seminars, and collaborative research with leading mathematicians.

  I have been accepted into and successfully completed the course Topology I, earning an **A+ grade (97.7/100)** Click here to view my transcript.

  The course covered:

  - Topological and metric spaces, manifolds, compactness, and continuity.
  - Classification of surfaces, Euler characteristic, and orientability.
  - Homotopy, covering spaces, and fundamental groups.
  - Infinite constructions and counterexamples in topology.
  - Introduction to knot theory and the Alexander–Conway polynomial.

Teaching Experience

- **Visiting Lecturer**
  Faculty of Technology, University of Colombo
  08 June 2026 – 19 June 2026
  - Coordinate and conduct a 10-day Intensive Mathematics lecture series for first-year undergraduate students.
- **Visiting Lecturer**
  GISM - Graduate Institute of Science and Management
  19 May 2026 – Present
  - Teaching the course FMS.011 Core Mathematics I
- **Teaching Assistant**
  Department of Mathematics, University of Sri Jayewardenepura
  02 June 2025 – 30 April 2026
  - Conducted MATLAB practical sessions for the course MAT 213 2.0 Numerical Methods, assisting students in implementing numerical methods and solving applied mathematical problems.
  - Led tutorial classes and supported students' understanding of key concepts in several core courses, including:
    - MAT 221 2.0 Functions of Several Variables
    - MAT 222 2.0 Partial Differential Equations
    - MAT 1052 Linear Algebra I
    - MAT 211 1.0 Linear Algebra II
    - MAT 212 2.0 Real Analysis I
    - MAT 376 3.0 Real Analysis II
  - Assisted in marking assignments, grading quizzes, and evaluating mid-semester examinations.
  - Lecturer in Charge of the course MAT 1100 Mathematical Software
    Department of Mathematics, University of Sri Jayewardenepura Nov 2025 – April 2026
    * Conducted lecture sessions covering essential mathematical software tools and computational techniques, mainly focusing on Maple software.
    * Prepared, evaluated, and discussed assignments, exercises, and end-semester examinations.
    * Conducted and managed tutorial sessions to support student learning.

Research Interests

Mathematical Biology, Computational Mathematics, Mathematical Modeling, Functional Analysis, Topology, Numerical Analysis, Probability Theory and Statistics.

Research Experience

Current Engagement in Ongoing Research:

- *Collaborating with Prof. Samuel T. Hess, Dr. Siyath Gunawardene and Dr. BPW Fernando* on a research project focusing on a mathematics, physics, and cell-biology inspired analysis of forecasting Hemagglutinin (HA) influenza protein trajectories in a cell membrane using filtering techniques specifically Kalman filtering. The aim is to investigate, within the framework of physics-inspired neural networks, determine the limit of noise encountered when predicting future states (noisy filtering), and to incorporate the Singer–Nicolson model to improve state-prediction accuracy. An additional goal is to examine whether the model should be re-formulated in light of new measurement data. Although we are still in the initial stage of problem identification, we are hopeful that, as the work progresses, we will be able to refine our methodology further and ultimately prepare the results for publication.

Completed Research and projects

- **Final Year Research Project:** *An Investigation of Fredholm Operators and Fredholm Index on Separable Hilbert Spaces (Achieved A+ Grade).*
  Click here to View My Thesis
  - Explored the deep connections between functional analysis and topology within a rigorous mathematical framework.
  - Studied separable Hilbert spaces.
  - Investigated compact operators, Fredholm operators, and the Fredholm index.
  - Analyzed the elegant connection between the Fredholm index and the topological invariant, the winding number.
- **Medical Diagnosis System Using ASP Prolog** *Course Project – MAT 358 2.0 An Introduction to Answer Set Prolog (Achieved A+ Grade)*
  - Collaborated in a group to develop a knowledge-based diagnostic system using ASP Prolog for identifying common illnesses based on user symptoms.
  - Implemented logical rules for over 20 medical conditions, including flu, food poisoning, heart attack, COVID-19, and diabetes.
  - Designed a scenario-based model simulating a clinic environment to evaluate real-time patient symptom input and generate diagnostic output.
  - Ensured system robustness by testing multiple patient cases under varying symptom combinations.
  - Delivered a comprehensive documentation report detailing rule formation, logical reasoning, and sample outputs.

Skills

- **Teaching and Mentoring:** Experiencing in university-level lecturing and conducting undergraduate tutorials, guiding students, and supporting learning in Mathematics.
- **Research and Analytical Skills:** Strong foundation in mathematical concepts with the ability to explore advanced topics and engage in research-oriented work.
- **Programming and Computational Tools:** MATLAB, Maple, R, SPSS, Minitab, Python, Java.
- **Typesetting:** Proficient in LaTeX for academic and educational documents.
- **Soft Skills:** Strong leadership, effective communication, and collaborative teamwork in academic environments.

Leadership Experience

Former Vice President, Mathematics Society *University of Sri Jayewardenepura, Sri Lanka* - (November 16, 2023 – July 11, 2024)

- Led the organization of the Japura Math Quiz 2024 (January 11, 2024)
- Led a two-day GCE O/L math seminar at M.D.H. Jayewardana Maha Vidyalaya, Battaramulla, in collaboration with its alumni (March 25–26, 2024).

References

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- Dr. R.P.K. De Silva
  Research Fellow,
  University of Réunion (Saint-Denis, Réunion, France) & The Research Centre for Quantum Information, Simulation and Algorithms (QUISA),
  University of Western Australia (Perth, Australia).

  Former Senior Lecturer (Gr. II),
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I hereby declare that the above information is true and accurate to the best of my knowledge and belief.